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A Bi-Directional Ge'ez-Amharic Neural Machine Translation:

a Deep Learning Approach

ABSTRACT

Currently, due to globalization, our world is moving into one village and human languages are being transnational. So far, human interpreters have been resolving communication gaps between two people who speak different languages. However, since human translation is costly and inconvenient, many kinds of research are being done to resolve this problem with Machine Translation (MT) techniques. MT is a process of automatically translating text or speech from one human language to another by computers. Neural Machine Translation uses Artificial Neural Networks such as Transformers, which are the state of the art models that shows promising result over the previous MT models. Several ancient scripts written in the Ge'ez language that need to be translated are available in Ethiopia and abroad. Currently, youth and researchers are interested to learn and involve in research areas of Ge'ez and Amharic manuscripts. This thesis, therefore, aims to demonstrate the capabilities of deep learning algorithms on MT tasks for those morphologically rich languages. A bi-directional text-based Ge'ez-Amharic MT was tested on two main different deep learning models viz. Seq2Seq with attention, and Transformer. A total of 20,745 parallel corpora was used for the experiment, from which the 13,787 parallel sentences were collected from former researchers and a new 6958 parallel corpus was prepared. In addition, a Ge'ez Latin numeric corpus having 3,078 parallel lines has been added to handle numeric translation. We conducted four experiments, and the transformer outperforms other techniques by scoring 22.9 and 29.7 BLEU scores from Ge'ez to Amharic and vice versa using 20,745 parallel corpora. The typical Seq2Seq model improves the BLEU score of the SMT model, obtained by previous researchers with BLEU scores of +0.65 and +0.79 that is 2.46% and 4.66% increment from Ge'ez to Amharic and from Amharic to Ge'ez using 13,833 parallel sentences. Doing further researches with clean larger corpus size and pre-trained models may improve the result we have reported in this work. However, we faced a scarcity of corpus and pre-trained models for Amharic and Ge'ez languages to get better results.

Keywords: Artificial Neural Network, Attention, BLEU, Seq2seq, Machine Translation, Neural Machine Translation, Transformer

I. Introduction

Machine Translation (MT) is an automatic translation of text or speech from one natural language to another, preserving the meaning of the input text, and producing fluent text in the target language [1, 2, and 3]. MT is a relatively old task that has taken a long journey in research and development. Over the years, two major MT approaches have emerged; a rule-based approach and the corpus-based approach. In the rule-based approach, experts' knowledge about the source and the target language to develop syntactic, semantic, and morphological rules is required to achieve the translation. Whereas, under the corpus-based approach, there is a parallel corpus¹ built by human experts from where the knowledge is automatically extracted by analyzing translation examples (bilingual text)² [2,8,9,11,12,13]. Corpus-based MT includes Example-Based MT (EBMT), Statistical Machine Translation (SMT), Neural Machine Translation (NMT), and other Hybrid MTs [2, 3, 4, 14 , 15]. These are discussed under section 2.4 in detail. SMT is an MT paradigm where translations are generated based on statistical models, whose parameters are derived from the analysis of bilingual text corpora. It is focusing on finding the translation with the highest probability of occurrence of words based on existing translations. Google launched Google Translate based on SMT in 2006. However, later on, after 10 years of research, Google substitute SMT with NMT to increase fluency and accuracy since SMT had poor grammatical accuracy [5]. Neural Machine Translation on the other hand is the newest approach to machine translation and is based on Artificial Neural Networks that consist of nodes conceptually modelled after the human brain. The complex and dynamic nature of networks in NMT allows the model to guess more educated and appropriate target text about the context. NMT systems iteratively learn and adjust weights to provide the best output than SMT but require a lot of processing power [6,16,17,18]. The relationship between AI, MT, and Deep Learning is depicted in the following figure.

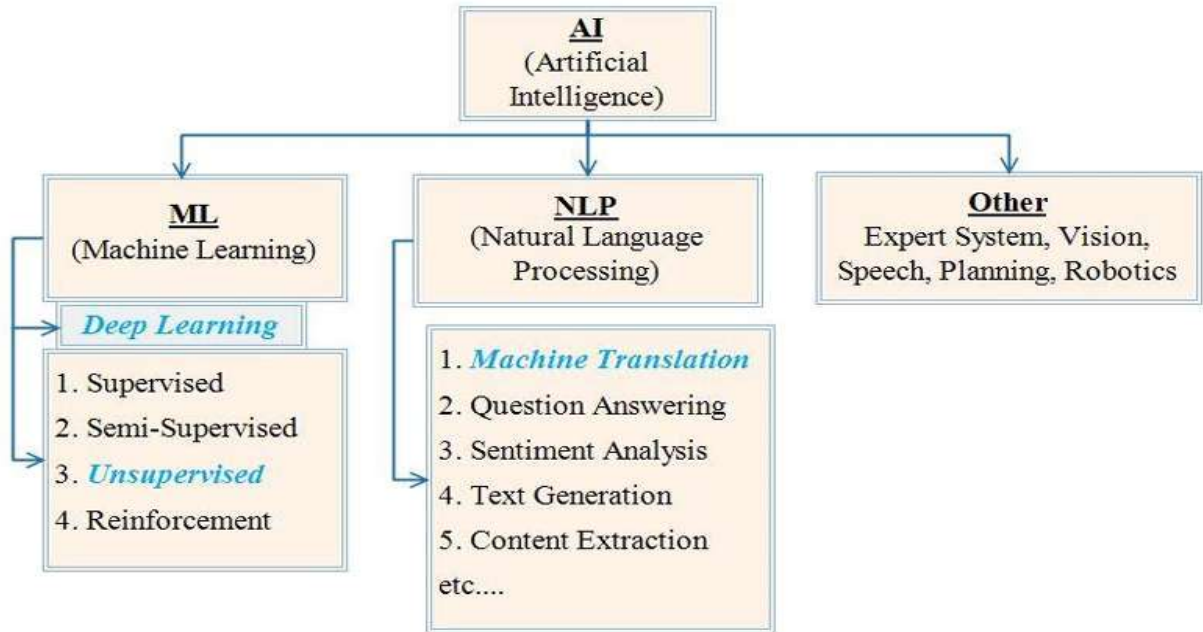


Fig 1. The relationship between AI, ML, and Deep Learning

II. Materials and method

Corpus Preparation, Implementation Tools and evaluation matrices:

We will use an existing domain-specific bilingual corpus having 13,787 parallel sentences prepared by the former researcher [2,19,21,23] from the bible, Mass, praise of Mary, and other Ethiopian Orthodox Church's faith books. We will also collect and prepare additional corpus in addition to the existing one. An experiment will be conducted with the formerly available corpus to determine whether the new NMT is effective for low-resourced languages and to be comparable with previous SMT results reported by [2,20,22]. However, The NMT will also be checked with both the existing corpora and with the existing plus the newly added corpus. In addition, parallel Ge'ez-Amharic numeral datasets will be added to handle numeric translation, Python will be used as a tool to develop models with Pytorch, and Open NMT will be used for confirmation. Python supports extensive collections of special libraries for implementing deep learning methodologies. Particularly offers a neater and faster way to build highly performing algorithms. Word embedding is a representation of words in the form of real-valued vectors, where words that have the same meaning have a similar representation. First, words are transformed into vocabulary (dictionary, a list of unique words with their corresponding indexes) then converted to one-hot encoding vector, finally to vectors of continuous real value numbers in a predefined vector space.

It normally involves a mathematic embedding from a high-dimensional sparse³³ vector space (e.g., one-hot encoding vector space, in which each word takes a dimension) to a lower-dimensional dense³⁴ vector space. Hence, embedding is a low-dimensional vector that captures a lot of syntactic and semantic information of words and their relationships. The dataset were collected from different source and the following figures indicates the sample dataset after passing the corpus and embedding challenges.

500	እስማኤል ውእቱ	500	እስማኤል ነው
501	ስምየ እስማኤል ውእቱ	501	ስሜ እስማኤል ነው
502	ስመ ዚአየ እስማኤል ይሰመይ	502	የኔ ስም እስማኤል ይባላል
503	አይቱ ውእቱ ዘመዳእኩ ዚአከ	503	እንተስ ከየት ነው የመጣኸው
504	አንሰ ዘመዳእኩ እምነ ኤርትራ ውእቱ	504	እኔስ የመጣሁት ከ ኤርትራ ነው
505	እምነ ኤርትራ ውእቱ ዘመዳእኩ	505	ከኤርትራ ነው የመጣሁጥ
506	ውስተ ኤርትራ እኔሉ	506	ኤርትራ ውስጥ እኖራለሁ
507	ተአምኖ ዘብእሲ ውብእሲት	507	የወንድና የሴት ሰላምታ
508	በሐከ እጉየ	508	እንዴት አደርክ ወንድሜ
509	እግዚአብሔር ይሰባሕ በሐኪ እጎትየ	509	እግዚአብሔር ይመስገን እንዴት አደርሻ እጎቴ
510	መኑ ስመክ እሁየ	510	የኔ ወንድም ስምህ ማን ይባላል
511	ኤልያስ እሰመይ	511	ኤልያስ እባላለሁ
512	ስመዚአየ ኤልያስ ውእቱ	512	የኔ ስም ኤልያስ ይባላል
513	ኤልያስ ይሰመይ ስምየ	513	ስሜ ኤልያስ ይባላል
514	ወመኑ ስመ ዚአኪ	514	ያንቺስ ስም ማን ነው
515	ኮከብ ውእቱ	515	ኮከብ ይባላል
516	አስቴር እሰመይ	516	አስቴር እባላለሁ
517	ስመዚአየ አስቴር ውእቱ	517	የኔ ስም አስቴር ነው
518	ኮከብ ይሰመይ ስምየ	518	ስሜ ኮከብ ይባላል
519	እስፍንት ውእቱ እድሜኪ	519	እድሜሽ ሰንት ነው
520	እስፍንት ውእቱ መዋዕልኪ	520	እድሜሽ ሰንት ነው
521	እስፍንት ውእቱ እድሜኪ ኮከብ	521	ኮከብ እድሜሽ ሰንት ነው
522	እስፍንት ውእቱ መዋዕልኪ አስቴር	522	እድሜሽ ሰንት ነው አስቴር
523	እድሜየ እስራ ወሀመስቱ ውእቱ	523	እድሜየ ሃያ አምስት ነው
524	እድሜየ ፳፭ ውእቱ	524	እድሜየ 25 ነው
525	ዘተወለድኩ በዐሥርቱ ወተስዓቱ ምዕት ተስዓ ወሰመንቱ ዓም	525	የተወለድኩት በአስራ ዘጠኝ መቶ ዘጠና ስምንት ዓም ነው
526	ዘተወለድኩ በ፲፱፻፲፯ ዓም	526	የተወለድኩት በ1989 ዓም ነው

Fig 2 Sample dataset from Ge'ez Amharic corpus in distinct files

1	0	0	358	፻፲፩	111	358	፻፲፩	111
2	አልቦ	ዜሮ	359	፻፲፩	111	359	፻፲፩	111
3	፩	1	360	ምዕት	ዐሠርቱ	ወእሐዱ	መቶ	አስራ አንድ
4	አሐዱ	አንድ	361	፻፲፪	112	361	፻፲፪	112
5	አሐደ	አንድ አደረገ	362	፻፲፪	112	362	፻፲፪	112
6	ዋሕድ	አንድ	363	ምዕት	ዐሠርቱ	ወክልኤቱ	መቶ	አስራ ሁለት
7	አሚሩ	መጀመሪያ	364	፻፲፫	113	364	፻፲፫	113
8	አሚሩ	አንደኛ	365	፻፲፫	113	365	፻፲፫	113
9	፩ይ	፩ኛ	366	ምዕት	ዐሠርቱ	ወሠለስቱ	መቶ	አስራ ሰባት
10	፩ይ	1ኛ	367	፻፲፬	114	367	፻፲፬	114
11	ቀዳማይ	ቀዳሚ	368	፻፲፬	114	368	፻፲፬	114
12	ቀዳማይ	አንደኛ	369	ምዕት	ዐሠርቱ	ወአርባዕቱ	መቶ	አስራ አራት
13	፪	2	370	፻፲፭	115	370	፻፲፭	115
14	ክልኤ	ሁለት	371	፻፲፭	115	371	፻፲፭	115
15	ክልኤቱ	ሁለት	372	ምዕት	ዐሠርቱ	ወገምስቱ	መቶ	አስራ አምስት
16	ክልኤተ	ሁለቱን	373	፻፲፮	116	373	፻፲፮	116
17	ክልዓይ	ሁለተኛ	374	፻፲፮	116	374	፻፲፮	116
18	ሠንዶ	ሁለተኛ	375	ምዕት	ዐሠርቱ	ወስድስቱ	መቶ	አስራ ስድስት
19	፪ይ	፪ኛ	376	፻፲፯	117	376	፻፲፯	117
20	፪ይ	2ኛ	377	፻፲፯	117	377	፻፲፯	117
21	፫	3	378	ምዕት	ዐሠርቱ	ወሰብዐቱ	መቶ	አስራ ሰባት
22	ሠለስቱ	ሰባት	379	፻፲፰	118	379	፻፲፰	118
23	ሠለስተ	ሰባቱ	380	፻፲፰	118	380	፻፲፰	118
24	ሠሉሱ	ሰባተኛ	381	ምዕት	ዐሠርቱ	ወስምንቱ	መቶ	አስራ ስምንት
25	ሣልሳይ	ሰባተኛ	382	፻፲፱	119	382	፻፲፱	119
26	፫ይ	፫ኛ	383	፻፲፱	119	383	፻፲፱	119
			384	ምዕት	ዐሠርቱ	ወተስዐቱ	መቶ	አስራ ዘጠኝ

Fig 3 Sample dataset the numeric corpus

Machine Translation Evaluation Metrics

There are generally two types of MT evaluation metrics, Human and automatic MT evaluation metrics [23,24,25]. A linguist that evaluates segments manually does human evaluation, which is intensive but expensive and time-consuming. Whereas Automatic evaluation of MT is the evaluation of translated contents using automated metrics such as BLEU, NIST, METEOR, and so on. Automated metrics emerged to address the need for objective, consistent, quick, and affordable assessment of MT output as opposed to a human evaluation. The most appropriate automatic metric for measuring one's MT system will depend on the language, content type, use case, and MT approach. The BLEU metric approach is applied for this research.

BLEU (Bi-Lingual Evaluation Understudy) is the most widely used metric for MT evaluation and was first proposed by Kishore Papineni, et al [26,27,28,2] in 2002. The idea behind BLEU is that “the closer the MT is to a linguist translation, the better it is”.

$$BLEU=BP.exp\sum_{n=1}^N WnLOGPn$$

Experimental program

For conducting an experiment, there are three different there three distinct corpus types and formats.

- The old corpus collected from the previous researcher [2] has a size of 13,787 parallel Ge'ez-Amharic sentences and was a domain-specific dataset, which is completely from the bible. However, the researcher reported as 13,833 parallel sentences were used in their work, but we got only 13,787 instead, and we added 46 religious parallel sentences from the new prepared one to be comparable with their work.
- The old pre-processed 13,787 corpus plus the new conversation corpus, has a size of 6,958 and makes the new one 20,745 parallel sentences
- The 20,745 parallel sentences with the addition of a new numeric corpus having 3,078 lines, that contains Ge'ez numerals in a different writing system and their corresponding Equivalent values in Latin number and Amharic text, which makes the new corpus 23,823. This numeral corpus was added to handle number translations.

Table 1. The different experiments, models, and corpus size used

No	Experiment	Model	Corpus size	Percentage Split
1	Experiment 1:	OpenNMT	13787(Old)	84%:8%:8% (T:T:V)
2	Experiment 2:	Transformer, OpenNMT	20,745(new)	80%:10%:10% (T:T:V)
3	Experiment 3:	Seq2Seq (LSTM with Attention)	20,745(new)	90%:10% (T:T)
4	Experiment 4:	Transformer, OpenNMT	23,823(new with numeric)	80%:10%:10% (T:T:V)

III. Experimental Result and discussion

The Amharic corpora have 20,745 sentences having 248, 114 tokens (words), with an average of 12 words per sentence which has 1,217,793 characters without line ending, 39437 Types, with the longest sentence having 142 words and the shortest sentence with 1 word. Similarly, the Ge'ez corpora have 20,745 sentences having 242,946 words, with an average of 12 words per sentence and 1,166,242 characters without line ending, with the longest sentence with 152 words and the shortest sentence with one word. The other corpus types and their experimental detail shown below in table 2.

Table 2. The length of sentence, token and type used for each experiment

	Total corpus					Percent age Split	Ge'ez			Amharic		
		Geez		Amharic								
	Sente nce	Toke n	Typ e	Toke n	Typ e		Senten ce	Toke n	Type	Sente nce	Toke n	Type
Experi ment 1	13,833	163,321	34,229	166,131	37,672	Training	11,067	129,828	29,601	11,067	132,468	32,584
		788,652 characters		819,908 characters		Testing	1,383	16,657	4,860	1,383	16,574	5,052
						Validati on	1,383	16,836	6,159	1,383	17,089	6,605
Experi ment 2	and 3 20,745	242,946	36,050	248,114	39,437	Training	16,596	201,936	34,983	16,596	205,235	38,345
		1,166,242 characters		1,217,793 characters		Testing	2,075	26,630	9,594	2,075	27,936	10,115
						Validati on	2,075	14,380	5,248	2,075	14,943	5,772
Experi ment 4	23,823	248,551	38,047	253,733	40,489	Training	19,059	199,885	35,974	19,059	203,184	38,184
		1,190,746		1,236,735		Testing	2,382	30,300	10,177	2,382	31,344	10,740
						Validati on	2,382	18,366	6,345	2,382	19,205	7,064

After conducting many experiments averagely, the result is shown in the table below.

Table 3. The final BLEU score result

Model	Corpus size	Ge'ez to Amharic	Amharic to Ge'ez	Hidden size of encoder and decoder	GPU Time	CPU Time
OpenNMT	13833(Old)	15.79	16.94	2	18m 8s	-
	20,745(new)	21.7	28.2	4	20m 25s	-
	23,823(new with numeric)	11.2	18.4	4	13m 54s	-
Sequential (LSTM with Attention)	20,745(new)	19.3	23.4	4 with 440 hidden neurons	30m 7s	10h 46m 33s
Transformer	20,745(new)	19.8	24.8	2	18m:13s:46ms	-
		22.9	29.7	4	21m:15s:52ms	-
	23,823(new with numeric)	16.5	20.1	4	28m:23s:12ms	-

IV. Conclusion

Many experiments different results were recorded for all translations with BLEU score metrics. In addition, questionnaires were prepared for Ge'ez and Amharic linguists to evaluate the translated texts between the two languages via the proposed models. In the automatic evaluation, the Transformer model achieved a higher result than the other models. Particularly, it improved the translation quality by 1.2 and 1.5 BLEU scores from Ge'ez to Amharic and vice versa over the OpenNMT model. Moreover, with the human manual evaluation, the transformer model outperforms the OpenNMT with 5.4% translation quality. Futurely, Applying semi-supervised approaches such as adding a monolingual corpus for each language can improve the translation quality.

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